

Bhabatosh Senapati

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EDUCATION

The Ohio State University

(Master of Computer Science & Engineering)

Columbus, OH

May 2025

Biju Patnaik University of Technology

(Bachelor of Technology in Mechanical Engineering)

Rourkela, India

May 2021

TECHNICAL SKILLS

Languages : Java, Go (Golang), Python, TypeScript, SQL, JavaScript (ES6)

Frameworks & Cloud: Spring Boot, Node.js, React, Next.js, GCP, Azure, Docker, Kubernetes, Ansible, Concord, Looper

Data & AI: PostgreSQL, Redis, Pinecone, OpenAI API, LangChain, Prisma ORM, Kafka, Jenkins

Core Concepts: Microservices, REST APIs, CI/CD, System Design, RAG Pipelines, WebSockets, Linux

PROFESSIONAL EXPERIENCE

Ontopharma Technologies

Software Engineer (Full Stack & AI)

Remote

Sep 2025 – Present

- Engineered an intelligent AI agent using **OpenAI GPT-4 function calling** to parse unstructured PDF data and automate FDA submission routing, achieving **85% accuracy** in classifying complex submission types (NDA, ANDA) and reducing manual review time by **30%**.
- Implemented a high-performance asynchronous ETL pipeline in **TypeScript** to ingest **500K+ clinical trials**, utilizing **Pinecone** vector storage and **LangChain** embeddings to power a semantic search RAG-based regulatory guidance system.
- Architected scalable full-stack workflows on **Next.js** and **React**, leveraging **Prisma ORM** and **PostgreSQL** to implement secure Server Actions, Role-Based Access Control (RBAC), and interactive drug composition analysis tools.
- Orchestrated** the deployment of containerized microservices on **GCP** using **Docker** and **Kubernetes**, automating multi-stage CI/CD pipelines via **Concord** and **Looper** to enable zero-downtime releases and reduce cycles by **40%**.

The Ohio State University (Prof. Rajiv Ramnath)

Research Intern

Columbus, OH

Jun 2024 – Dec 2024

- Designed** a resilient spatio-temporal knowledge graph using **Neo4j** on **Azure** for a ****911 Emergency Response System****, processing **10K+ records** to identify congestion patterns and optimize ambulance arrival times by **15%**.
- Developed a Dockerized Python NLP pipeline** to parse ****unstructured distress logs****, implementing **Named Entity Recognition (NER)** to pinpoint incident locations and improve routing logic precision.
- Applied** graph traversal algorithms to calculate optimal paths in real-time, dynamically accounting for urban obstacles like road closures to enhance ****critical life-safety dispatch**** decisions.

Niv-Tech Solutions Private Limited

Software Engineer (Fintech & Payments Platform)

Bhubaneswar, India

Jun 2021 – Jul 2023

- Engineered** a high-volume **Financial Transaction Engine** using **Java Microservices**, optimizing multi-threaded processing to handle **\$500k+** in monthly settlements with strict **ACID compliance**, ensuring zero ledger discrepancies across distributed databases.
- Spearheaded** the strategic migration of a legacy monolithic core banking application to **Dockerized Microservices**, automating configuration management and infrastructure provisioning via **Ansible** playbooks on **Linux** servers to enhance fault isolation.
- Developed** resilient integration layers for multiple payment gateways (**PayPal, Razorpay, PhonePe, GPay**) using **Circuit Breakers** and **Idempotency keys**, standardizing diverse API responses and preventing double-charges during intermittent network failures.
- Secured** sensitive Payment Card Industry (PCI) data by implementing robust **Tokenization** strategies and **OAuth 2.0** with **Azure AD**, enforcing encryption standards to reduce fraud risk while improving user login success rates by **73%**.

Indian Institute of Technology Guwahati (IIT G)

Research Intern (IoT & Data Systems)

Guwahati, India

May 2020 – Jan 2021

- Architected** a real-time **IoT Data Pipeline** for high-frequency sensor telemetry, utilizing **LabVIEW** to ingest and process multi-stream data for automated system analysis.
- Developed** algorithmic control logic for Electronic Control Units (ECUs), optimizing injection timing to achieve a **38% thermal efficiency improvement** through data-driven tuning.

PROJECTS

Coduter.com (Founder) | Go (Golang), WebSockets, Redis, Distributed Systems | [Live Link: coduter.com](#)

Aug 2024 – Dec 2024

- Founded** and engineered a **real-time global competitive coding platform** (akin to Chess.com), implementing a **Matchmaking Service** in **Go** to pair coding enthusiasts for 1v1 timed algorithmic battles.
- Synchronized** live game state across distributed clients using **WebSockets** and **Redis Pub/Sub**, ensuring sub-50ms latency for code execution and timer updates.
- Implemented** a custom heartbeat mechanism to detect client disconnections and handle automatic reconnections, ensuring a seamless competitive experience.

Scalable E-commerce Inventory Engine | Java, Spring Boot, Kafka, Redis, PostgreSQL

Jan 2024 – Apr 2024

- Architected** a high-availability inventory microservice handling concurrent order reservations during simulated flash sales.
- Implemented** a distributed locking mechanism using **Redis** (Redlock) to prevent overselling of low-stock items, ensuring 100% inventory accuracy.
- Designed** an event-driven order processing pipeline using **Kafka**, decoupling stock deduction from payment confirmation to improve checkout latency by **40%**.

LEADERSHIP & INTERESTS

Leadership: Co-Founder of Coduter (Led product vision and engineering); Mentored 5+ junior developers on Distributed Systems concepts.

Interests: Real-time Event Architectures, System Design, Open Source Contribution, Physical Fitness & Weight Training.